

Amir H. Karimi

ASSISTANT PROFESSOR | O'DONOVAN CHAIR IN TRUSTWORTHY AI | MACHINE LEARNING SCIENTIST & ENGINEER

✉ amirhkarimi@gmail.com | 🌐 www.amirhkarimi.com | 🐦 amirhkarimi_ | 📄 Google Scholar

My career spans **top academic institutions** (Toronto, Waterloo, Stanford, ETH Zürich, Max Planck), **leading industry labs** (Google Brain, DeepMind, Meta AI), **major tech firms** (Meta, BlackBerry), and **startup ecosystems** (NEXT AI, The Next 36). This rare breadth gives me a deep appreciation for practical challenges, while my technical grounding and collaborations with world-class teams enable me to develop solutions that are both theoretically rigorous and real-world ready. My work focuses on explainable, trustworthy AI, and has been recognized with honors across research, industry, and teaching.

Employment

University of Waterloo

ASSISTANT PROFESSOR OF MACHINE LEARNING

Waterloo, CA

Sep 2023 - p.

- Tenure-track Assistant Professor in the Electrical and Computer Engineering Department & School of Computer Science (cross-app.)
- PI of the **Collaborative Human-AI Reasoning Machines (CHARM) Lab**, dedicated to advancing safe and trustworthy human-AI teams
- **Affiliations:** Vector Institute Faculty Affiliate, Future of Life Institute Member of AI Safety Community Researchers

Google DeepMind

RESEARCH SCIENTIST INTERN

London, UK

May 2022 - Oct 2022

- Improved search via type- & IO-based neurally-guided program synthesis. Mentors: Lars Buesing, David Amos, Jessica Hamrick

Google Brain

RESEARCH SCIENTIST INTERN

Waterloo, CA

Dec 2021 - Apr 2022

- Investigated the causal effect of training hyperparameters on ML explanation. Mentors: Been Kim, Simon Kornblith
- Successfully published at ICML 2023, under the title “On the Relationship Between Explanation and Prediction: A Causal View” [C16]

Meta (Facebook) Inc.

SOFTWARE ENGINEER

New York, USA

Aug 2015 - Sep 2016

- Full-stack software engineer on the Enterprise Eng. team, responsible for front-end dev using React and hphp among others
- Implemented the landing page, a customizable notification framework, and the testing and logging platform for the Org Tool
- Successfully published at EMNLP 2016, under the title “Key-Value Memory Networks for Directly Reading Documents” [C2, P1]

Meta (Facebook) Inc.

SOFTWARE ENGINEER INTERN

Menlo Park, USA

Feb 2014 - Apr 2014

- Delivered a first-of-a-kind reporting tool for Facebook's largest Business Manager ad clients to view and manage historical ad budgets.
- Successfully reduced TTI < 7sec for Facebook's largest ad clients (100K+ ad accounts), by optimizing front-end JavaScript rendering.

BlackBerry Inc.

SOFTWARE ENGINEER INTERN

Toronto, CA

May 2013 - Dec 2013

- Implemented an automation framework to test 6,000+ ported selenium webdriver and WebKit tests for BB10.
- Successfully integrated browser team's test automation results with company central test database.

Stanford University

UNDERGRADUATE RESEARCH ASSISTANT - HIGH-FREQUENCY LAB

Stanford, USA

May 2012 - Aug. 2012

- Developed a helical antenna and used machine learning for precise 2D localization enabling hand gesture recognition.

Education

ETH Zürich & Max Planck Institute for Intelligent Systems

PHD IN COMPUTER SCIENCE

Zürich, CH & Tübingen, DE

Oct 2018 - Jul 2023

- **Thesis:** “Advances in Algorithmic Recourse: Ensuring Causal Consistency, Fairness, & Robustness”
- **Supervisors:** Prof. Bernhard Schölkopf & Prof. Isabel Valera
- **Major Awards:** NSERC CGS-D PhD Fellowship, Max Planck ETH PhD Fellowship, Google PhD Fellowship, ETH Zurich Medal

University of Waterloo

MMATH IN COMPUTER SCIENCE

Waterloo, CA

Sep 2016 - Apr 2018

- **Thesis:** “Exploring New Forms of Random Projections for Dimensionality Reduction”
- **Supervisors:** Prof. Alexander Wong & Prof. Ali Ghodsi
- **Major Award:** *Alumni Gold Medal* for highest standing across all master's programs at UWaterloo

University of Toronto

B.A.SC. IN ENGINEERING SCIENCE – ELECTRICAL AND COMPUTER STREAM

Toronto, CA

Sep 2010 - Jun 2015

- **Thesis:** “Benchmarking a Neuro-biologically Inspired Adaptive Controller”
- **Supervisors:** Prof. Chris Eliasmith & Prof. Richard Zemel
- **Major Award:** *Spirit of Engineering Science Award* for outstanding community contribution

Publications

My scholarly contributions on trustworthy, explainable, and causally-grounded AI have been **showcased almost exclusively at top-tier AI and ML venues**—including NeurIPS, ICML, AAAI, AISTATS, ACM FAccT, and ACM AIES—where peer-reviewed conference proceedings are the primary venue for high-impact dissemination for artificial intelligence research. I have authored influential works such as a comprehensive survey in the prestigious ACM Computing Surveys, contributed a book chapter, and hold a U.S. patent. My research on algorithmic recourse has significantly shaped the field of responsible AI, helping elevate it from an emerging topic to a formal policy criterion—now mandated in Canada’s Treasury Board Directive on Automated Decision-Making. **Several of my papers have each been cited over 100 times** and have been presented as spotlight and oral talks at top venues, reflecting sustained scholarly and policy impact. (📖 Google Scholar)

SPOTLIGHT (📖) ORAL (🎤) ≥ 100 CITATIONS (☆) BEST PAPER (🏆) PATENT (🔬) BOOK CHAPTER (📖) EQUAL CONTRIBUTION (*)

H-INDEX	G-INDEX	MAX CITATIONS	TOTAL CITATIONS	🎤	📖	☆	🏆	🔬	📖	*
16	34	1,384 [C2]	3,925	12	3	7	1	1	1	1

Patents

P1 🔬 2018 US Patent
-
“Key-Value Memory Networks”
Miller, Fisch, Dodge, **Karimi**, Bordes, Weston

Book Chapters

B1 📖 2022 Springer LNAI
-
“Towards Causal Algorithmic Recourse”
Karimi,* von Kügelgen,* Schölkopf, Valera

Journal Proceedings

J4 - 2026 Trends in CogSci
I.F. 17.2
Johnson, **Karimi**, Bengio, Chater, Gerstenberg, Larson, Levine, Mitchell, Rahwan, Schölkopf, Grossmann
J3 - 2025 Nature Scientific Reports
I.F. 3.9
“Temporal Convolutional Transformer for EEG Based Motor Imagery Decoding”
Altaheri, Karray, **Karimi**
J2 ☆ 2022 ACM Computing Surveys
I.F. 28.0
“A survey of algorithmic recourse: contrastive explanations & consequential ...”
Karimi, Barthe, Schölkopf, Valera
J1 - 2011 Optics Express
I.F. 3.2
“Automated detection and density assessment of keratocytes in the human ...”
Karimi, Wong, Bizheva

Conference Proceedings

C20 - 2026 COLM
A.R. ≤ %30.0
“How well can a large language model infer the value representations ...”
Ghorbansabagh, JI, **Karimi**, Grossmann,
C19 - 2026 ACM FAccT
A.R. ≤ %30.0
“Your Recourse, My Loss? Algorithmic Recourse under Shared Constraints”
Khotanlou, Larson, **Karimi**
C18 - 2024 ICML
A.R. %27.5
“Prospector Heads: Generalized Feature Attribution for Large Models & Data”
Machiraju, Derry, Desai, Guha, **Karimi**, Zou, Altman, Ré, Mallick
C17 🎤 2023 AAAI
A.R. %23.75
“Causal Adversarial Perturbations for Individual Fairness and Robustness in ...”
Ehyaei, Mohammadi, **Karimi**, Samadi, Farnadi
C16 - 2023 ICML
A.R. %27.9
“On the Relationship Between Explanation and Prediction: A Causal View”
Karimi, Muandet, Kornblith, Schölkopf, Kim
C15 - 2023 ICML
A.R. %27.9
“On Data Manifolds Entailed by Structural Causal Models”
Dominguez-Olmedo, **Karimi**, Arvanitidis, Schölkopf
C14 - 2023 ACM FAccT
A.R. %24.6
“Robustness Implies Fairness in Causal Algorithmic Recourse”
Ehyaei, **Karimi**, Schölkopf, Maghsudi
C13 ☆ 📖 2022 ICML
A.R. %21.9
“On the Robustness of Causal Algorithmic Recourse”
Dominguez-Olmedo, **Karimi**, Schölkopf
C12 ☆ 🎤 2022 AAAI
A.R. %15.0
“On the Fairness of Causal Algorithmic Recourse”
von Kügelgen, **Karimi**, Bhatt, Valera, Weller, Schölkopf
C11 🎤 2021 ACM-AIES
A.R. %38.0
“Scaling Guarantees for Nearest Counterfactual Explanations”
Mohammadi, **Karimi**, Barthe, Valera

C10	☆💡	2021	ACM FAccT A.R. %25.0	“Algorithmic Recourse: from Counterfactual Explanations to Interventions” Karimi , Schölkopf, Valera
C9	☆💡	2020	NeurIPS A.R. %20.1	“Algorithmic recourse under imperfect causal knowledge: a probabilistic ...” Karimi ,* von Kügelgen,* Schölkopf, Valera
C8	☆🎙️	2019	AISTATS A.R. %32.4	“Model-Agnostic Counterfactual Explanations for Consequential Decisions” Karimi , Barthe, Balle, Valera
C7	🎙️	2018	IJCNN A.R. %22.7	“Distance Correlation Autoencoder” Wang, Karimi , Ghodsi
C6	🎙️	2018	CVIS A.R. ≤ %40.0	“FEELS: a full-spectrum enhanced emotion learning system for assisting ...” Karimi ,* Boroomand,* Pfisterer, Wong
C5	🏆🎙️	2017	CVIS A.R. ≤ %40.0	“Ensembles of Random Projections for Nonlinear Dimensionality Reduction” Karimi , Shafiee, Ghodsi, Wong
C4	-	2017	CCN A.R. ≤ %40.0	“Synthesizing Deep Neural Network Architectures using Biological Synaptic ...” Karimi , Shafiee, Ghodsi, Wong
C3	🎙️	2017	ICIAR A.R. ≤ %40.0	“Discovery Radiomics via a Mixture Sequencers for Multi-Parametric MRI ...” Karimi , Chung, Shafiee, Khalvati, Haider, Ghodsi, Wong
C2	☆🎙️	2016	EMNLP A.R. %24.3	“Key-Value Memory Networks for Directly Reading Documents” Miller, Fisch, Dodge, Karimi , Bordes, Weston
C1	🎙️	2016	ICIP A.R. ≤ %40.0	“Spatio-temporal saliency detection using abstracted fully-connected ...” Karimi , Shafiee, Scharfenberger, BenDaya, Haider, Talukdar, Clausi, Wong

Workshop Proceedings

W4	🎙️	2026	ICML CTB	“A Unified Perturbation Framework for Analyzing Leaderboard Stability and ...” Oyarhoseini, Lin, Karimi
W3	🎙️	2025	EurIPS THEORY OF XAI	“Explainable AI is Causal Discovery in Disguise” Karimi
W2	-	2018	NeurIPS -	“Deep Variational Sufficient Dimensionality Reduction” Banijamali, Karimi , Ghodsi
W1	-	2017	NeurIPS -	“JADE: Joint Autoencoders for Dis-Entanglement” Karimi ,* Banijamali,* Ghodsi, Wong

Selected Pre-prints (as lead, senior, or core contributing author)

UP3	-	2026	-	“RecourseBench: A Modular Framework for Reproducible Algorithmic ...” Khotanlou, Ahmed, Tan, Abdelaal, Karimi
UP2	-	2026	-	“Metacognition Should Be the Scientific Framework for Bounded and ...” Ji, Grossmann, Karimi
UP1	-	2025	-	“Bridging Brain with Foundation Models through Self-Supervised Learning” Altaheri, Karray, Islam, Raju, Karimi

Advising

One of my greatest privileges as an advisor is witnessing students grow into independent researchers. I have mentored and trained **32 highly-qualified personnel** (27 as faculty) in a diverse team—across culture, gender, and seniority—of Postdoctoral, PhD, Master’s, and undergraduate students, supporting their growth through interdisciplinary research and global collaboration. Several have gone on to top PhD programs, fellowships, and roles at leading institutions.

ADVISEE	ONGOING	COMPLETED	TOTAL
Postdoctoral Fellows	1	2	3
PhD students	1	-	1
Master’s (thesis- /course-based)	2/-	3/5	5/5
Undergraduate (research/capstone)	2/-	5/6	7/6
OVERALL	6	21	27

Postdoctoral

2026-p.	Co-supervisor	Eugene Yu (PSYCH)	📅 UP2, C20	🏠 Waterloo	-
2026	Supervisor	Mohammad Hadi Sepanj (ECE)	-	🏠 Waterloo	-
2024-5	Co-supervisor	Hamdi Altaheri (ECE)	📅 J3, UP1	🏠 Waterloo	→ King Saud University

PhD

2026-p.	Supervisor	Ahmed Abdelaal	📅 UP3	🏠 Waterloo	-
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Master’s (thesis-based)

AUGUST 11, 2026

AMIR H. KARIMI · CURRICULUM VITÆ

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2025-p.	Supervisor	Hashir Ahmed (ECE)	UP3	Waterloo	-
2025-p.	Supervisor	Chenghao Tan (ECE)	UP3	Waterloo	-
2024-6	Supervisor	Zahra Khotanlou (ECE)	C19, UP3	Waterloo	-
2024-6	Co-supervisor	Maryam Ghorbansabagh (ECE)	C20	Waterloo	-
2025-6	Co-supervisor	Hosna Oyarhoseini (CS) (Vector Scholarship in AI)	W4	Waterloo	-

Bachelors (research)

2025-6	Supervisor	Farzan Mirshekari (ECE)	-	Waterloo	-
2025-6	Supervisor	Tom Wielemaker (MECH)	-	Waterloo	-
2025-6	Supervisor	Hamza Mostafa (ECE)	-	Waterloo	→ Open AI Inc.
2024	Supervisor	Abubakar Bello (ECE)	-	Waterloo	→ Microsoft Inc.
2024	Supervisor	Mohammadreza Alavi (ECE)	-	Sharif	-

Honors, Funding, & Awards

As an early-career researcher, I have been fortunate to receive strong support for my work—**securing over CAD \$1,250,000** in competitive funding from generous sponsors including the University of Waterloo, Waterloo.AI, NSERC, Google, and CIFAR. I am honored to have **received several highly competitive national and international recognitions**, listed below.

As a faculty

2025-p.	Sole-PI	O'Donovan Chair in Trustworthy AI	(CAD \$500,000)	Waterloo, ON
2025-6	Lead-PI	Mitacs Accelerate Entrepreneur Grant	(CAD \$90,000) 50%	Waterloo, ON
2025-6	Co-PI	CIFAR Catalyst Grant	(CAD \$100,000 33%)	Waterloo, ON
2024	Co-PI	Waterloo.AI Nexus of Data & AI Seed Funding	(CAD \$20,000 50%)	Waterloo, ON
2024-8	Sole-PI	NSERC Discovery Grant (highest amount awarded to an early-career researcher in Waterloo Engineering in 2024)	(CAD \$235,000)	Waterloo, ON
2024-8	Sole-PI	NSERC Discovery Grant Supplements	(CAD \$12,500)	Waterloo, ON
2024	Recipient	Igor Ivkovic Teaching Excellence Award (1 of 1; competitive, student-nominated teaching award received in my first term)	-	Waterloo, ON
2023	Recipient	Institution Startup Fund	(CAD \$170,000)	Waterloo, ON

As a student

2023	Recipient	ETH Zurich Medal for Outstanding Doctoral Performance	(CHF 2,000)	Zürich, CH
2021	Recipient	Google PhD Fellowship in AI for Social Good (1 of 17 globally)	(USD \$210,000)	Tübingen, DE
2018	Recipient	Max Planck ETH Center for Learning Systems PhD Fellowship	-	Tübingen, DE
2018	Recipient	NSERC Postgraduate Scholarship - Doctorate (PGS-D)	(CAD \$63,000)	Waterloo, CA
2018	Recipient	NSERC Canada Graduate Scholarship - Doctorate (CGS-D)	(CAD \$105,000) DECLINED	Waterloo, CA
2018	Recipient	President's Graduate Scholarship (PGS)	(CAD \$35,000) DECLINED	Waterloo, CA
2018	Recipient	David R. Cheriton Graduate Scholarship	(CAD \$20,000) DECLINED	Waterloo, CA
2018	Recipient	Alumni Gold Medal – UWaterloo's top master's award (1 of 1)	-	Waterloo, CA
2015	Recipient	Spirit of EngSci Award for exemplary non-academic impact	-	Toronto, CA

Teaching

Teaching is one of my greatest joys. I have had the privilege of educating diverse audiences—from (under)graduate students across Canada, Germany, and Switzerland to **over 40,000 learners online**—on topics such as machine learning and AI ethics. As a faculty member, I have taught **over 750 students in person** and was honored to receive the competitive, **student-nominated Igor Ivkovic Teaching Excellence Award** in my **first teaching term** at the University of Waterloo.

2026	Instructor	AI-CTO Executive Education (with WatSPEED)	~ 30 students	Waterloo, CA
2026	Co-instructor	Agents in the Wild: Building AI That Actually Does Things	~ 50 participants	Waterloo, CA
2024-26	Instructor	Introduction to Machine Learning (grad.)	~ 125–150 students	Waterloo, CA
2024-26	Instructor	Foundations of Computational Intelligence (ugrad.) (Igor Ivkovic Teaching Excellence Award , 2024)	~ 75–100 students	Waterloo, CA
2022-p.	Co-instructor	Providing free public education on basic & advanced AI subjects on YouTube, Instagram, Substack, & Medium	≥ 40,000 students	online

Invited Talks (Selected)

“Building Bridges: Towards Trustworthy Human-AI Decision Making” NSERC CREATE: Responsible Development of AI (keynote, 2026), Vector Institute (2024), Toronto Machine Learning Summit (2024), Waterloo Alumni Reunions (2024, 2025, 2026), Waterloo.AI Seminars (2024), Waterloo.AI Nexus of Data & AI Event (2024), Waterloo VIP Lab (2024), Waterloo ECE Seminars (2024)

“Explainable AI is Causality in Disguise” Apple Workshop on Reasoning and Planning (2026), EurlPS Theory of Explainable AI (2025)

“Algorithmic recourse: from theory to practice” Google Brain LUMA team (2022), Google DeepMind (2022), MILA (2022), IMS Annual Meeting (2022), Harvard University (2021), NEC Europe Labs (2021), Cyber Valley Health (2021), ETH IML Seminars (2020), UCL Causality Group (2020)

Panelist:

2026	The Canadian AI Journalism Network	~ 50 attendees	Canada
2024	Tech Horizons Executive Forum Panel on Trust in AI	~ 300 attendees	Toronto, CA
2024	BBC Radio 4 Episode on AI for Emotion Detection	-	online
2023	AI and the Transformation of Social Science Research	~ 50 attendees	Waterloo, CA

Scientific Reviewing

	Program Committee	Journal Reviewer	Grant Reviewer
2026	TMLR	ACM Transactions on Knowledge Discovery from Data (TKDD)	-
2025	ICLR, NeurIPS, AISTATS	Journal of Artificial Intelligence	NSERC Mitacs Accelerate
2024	-	-	NSERC Discovery Grants
2023	ICLR	-	-
2022	ICML, AISTATS	-	-
2021	ICLR, NeurIPS	Journal of Machine Learning Research	-
2020	ICLR, NeurIPS, ACM FAccT, ECML, AISTATS	-	-

Other Service

Workshop & Symposium Organization

2026	Co-organizer	ICLR Workshop on Unifying Concept Representation Learning	~ 50 attendees	Rio, Brazil
2026	Organizer	AI Workshop on “Agents in the Wild”	~ 50 attendees	Waterloo, CA
2026	Moderator	Society of Professional Engineers Panel on Trustworthy AI	~ 100 attendees	Waterloo, CA
2023	Co-organizer	ICML Workshop on Counterfactuals in Minds & Machines	~ 50 attendees	Hawaii, USA
2021	Co-organizer	ELLIS Workshop on Causethical ML	~ 50 attendees	online
2021	Co-organizer	ICML Workshop on Algorithmic Recourse	~ 50 attendees	online
2020	Co-organizer	NeurIPS Symposium on Algorithmic Recourse	~ 100 attendees	online
2020	Support	Machine Learning Summer School (MLSS)	~ 150 attendees	online

Program & Fellowship Selection

2024-5	Reviewer	ICML Workshops Proposal Review Committee	//	online
2024-p.	Reviewer	Vector Scholarship in AI Review Committee	///	Toronto, CA

Departmental & Faculty Service

2026-p.	Examiner	Engineering Capstone Symposium	/	Waterloo, CA
2024-p.	Dept. Repr.	Engineering Faculty Council	///	Waterloo, CA

Graduate Examination & Committee Service

2024-p.	Chair	PhD Comprehensive Exam	//	Waterloo, CA
2024-p.	Examiner	PhD Defence	/	Waterloo, CA
2024-p.	Examiner	PhD Comprehensive Seminar	/	Waterloo, CA
2024-p.	Examiner	PhD Comprehensive Proposal Exam	###	Waterloo, CA
2024-p.	Examiner	PhD Comprehensive Background Exam	### ///	Waterloo, CA
2024-p.	Examiner	Master’s Seminar/Thesis	///	Waterloo, CA